

Phishing and Malicious Email Classification Using LSTM

Deep Learning Applications – ELA-1
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1. Abstract

Email inboxes receive a large volume of legitimate and unwanted messages. Automatically identifying malicious or spam-like messages can support safer and more efficient inbox management. This project develops a Deep Learning pipeline using a Long Short-Term Memory (LSTM) network to classify raw email text into two categories: **Legitimate** and **Spam/Malicious**. The CEAS_08 email dataset was used, with the email subject and body combined as the model input. After text cleaning, tokenisation and padding, the data was supplied to an Embedding–LSTM classifier. The trained model achieved **99.21% test accuracy** on 7,214 test emails.

2. Dataset

Item	Description
Dataset	CEAS_08 email dataset
Total emails	39,154 (before removal of malformed rows/empty text during loading and preparation)
Input features	Email subject + email body
Target	Binary email label
Classes used	0 – Legitimate; 1 – Spam/Malicious
Train/Test split	80% training, 20% testing
Test samples	7,214

3. Data Preprocessing Workflow

The **subject** and **body** fields were combined into one text input. Missing subject/body values were replaced with empty strings. Empty emails were removed. Text cleaning converted content to lowercase, replaced URLs with a special URL token, removed HTML tags and non-alphabetic characters, and normalised extra whitespace. The cleaned text was then converted to integer sequences using a Keras tokenizer. The vocabulary was limited to 10,000 words and an out-of-vocabulary token was used for unseen words. All sequences were padded or truncated to a fixed length of 200 tokens.

Workflow: Raw Email → Subject + Body → Cleaning → Train/Test Split → Tokenisation → Padding/Truncation → LSTM Classification

4. Preprocessing Parameters

Parameter	Value
Vocabulary size	10,000
Maximum sequence length	200
Padding	Post-padding
Truncation	Post-truncation
Split	80% train / 20% test

Model Architecture and Training

5. LSTM Model Architecture

INPUT	Email Subject + Body
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PREPROCESSING	Cleaning → Tokenisation → Padding (200 tokens)
↓	
EMBEDDING	10,000-word vocabulary → 128-dimensional embeddings
↓	
LSTM	64 units
↓	
DROPOUT	0.5
↓	
DENSE	32 units, ReLU activation
↓	
DROPOUT	0.3
↓	
OUTPUT	1 unit, Sigmoid activation
↓	
CLASSIFICATION	Legitimate / Spam-Malicious

6. Architecture Explanation

Embedding layer: transforms token IDs into dense numerical vectors so that words can be represented in a form the neural network can learn from. **LSTM layer:** processes the token sequence and learns useful contextual patterns in email text. **Dropout:** randomly drops a fraction of activations during training to reduce overfitting. **Dense layer:** learns higher-level classification patterns. **Sigmoid output:** produces a value between 0 and 1 for the binary decision.

7. Hyperparameters

Hyperparameter	Selected value
Vocabulary size	10,000
Embedding dimension	128
LSTM units	64
Dense units	32
Dense activation	ReLU
Dropout	0.5 and 0.3
Output activation	Sigmoid
Optimizer	Adam
Loss function	Binary Cross-Entropy

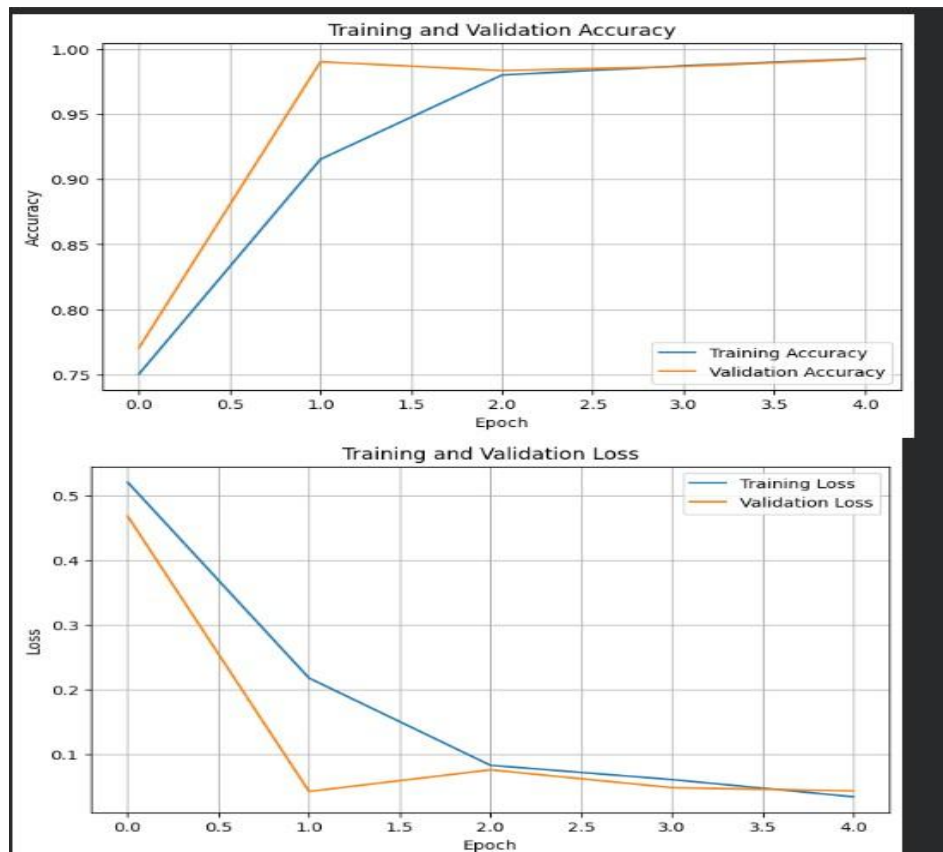
Batch size	64
Epochs	5

8. Training Procedure

The model was trained for 5 epochs using a batch size of 64. The Adam optimizer was used with binary cross-entropy as the loss function. Twenty percent of the training portion was used as a validation split during model training. The final model was evaluated on the held-out test set of 7,214 emails.

Evaluation, Results and Conclusion

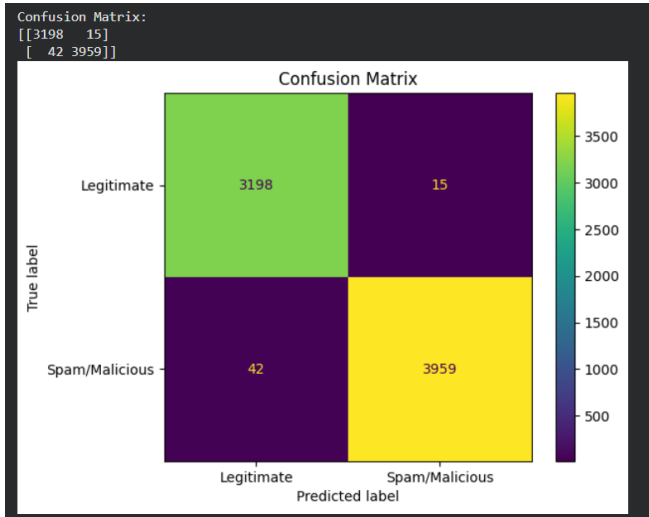
9. Training and Validation Curves



The accuracy curve rises to approximately 99% for both training and validation data, while the loss decreases substantially over the five epochs. The close final training and validation curves indicate that the model learned the classification patterns effectively without a large final train-validation gap.

10. Confusion Matrix

	Predicted Legitimate	Predicted Spam/Malicious
Actual Legitimate	3,198 (TN)	15 (FP)
Actual Spam/Malicious	42 (FN)	3,959 (TP)



11. Evaluation Metrics

Metric	Legitimate	Spam/Malicious
Precision	0.99	1.00
Recall	1.00	0.99
F1-score	0.99	0.99
Support	3,213	4,001

Overall test accuracy: 99.21% Test loss: approximately 0.0481

12. Conclusion

The developed LSTM-based pipeline successfully classifies email text into legitimate and spam/malicious categories. The combination of text preprocessing, tokenisation, padding, word embeddings and LSTM sequence modelling enabled the system to achieve **99.21% test accuracy** with an F1-score of approximately **0.99** for both classes. The confusion matrix shows a small number of misclassifications compared with the total test set. The results demonstrate that an LSTM can provide an effective baseline Deep Learning solution for automated email inbox classification. Because the CEAS_08 dataset is a spam/ham-oriented dataset, the malicious class should be interpreted as spam/malicious rather than as exclusively confirmed phishing.